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1 USGS/MDBC PCR Protocol 18S rRNA V9 (Lane-Medlin)

***WARNING: DETAILS BELOW HAVE NOT BEEN UPDATED.
PLEASE USE ABOVE VALUES IN YAML FRONT MATTER.***

1.1 PROTOCOL INFORMATION

1.1.1 Minimum Information about an Omics Protocol (MIOP)

- MIOP terms are listed in the YAML frontmatter of this page.
- See <https://github.com/BeBOP-OBON/miop/blob/main/model/schema/terms.yaml> for list and definitions.

1.1.2 Making eDNA FAIR (FAIRe)

- FAIRe terms are listed in the YAML frontmatter of this page.
- See <https://fair-edna.github.io/download.html> for the FAIRe checklist and more information.

- See <https://fair-edna.github.io/guidelines.html#missing-values> for guidelines on missing values that can be used for missing FAIR or MIOP terms.

1.1.3 Authors

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1.2 RELATED PROTOCOLS

PROTOCOL NAME	LINK	VERSION	RELEASE DATE
Not provided			

1.3 RELATED EXTERNAL PROTOCOLS

PROTOCOL NAME	LINK	ISSUER / AUTHOR	ACCESS DATE
Not provided			

1.3.1 Protocol Revision Record

VERSION	RELEASE DATE	DESCRIPTION OF REVISIONS
1.0.0	2025-12-15	Initial release
1.0.1	2025-12-15	Updated YAML front matter - note that details in main text still reflect AOML parameters and have not been updated

VERSION	RELEASE DATE	DESCRIPTION OF REVISIONS
1.0.2	2025-12-17	Updated entire protocol to reflect USGS parameters

1.3.2 Acronyms and Abbreviations

ACRONYM / ABBREVIATION	DEFINITION
USGS	United States Geological Survey
NOAA	National Oceanographic and Atmospheric Administration
MSU	Mississippi State University
NGI	Northern Gulf Institute
PCR	Polymerase chain reaction
eDNA	environmental DNA
NTC	No template control
EtOH	Ethanol

1.3.3 Glossary

SPECIALISED TERM	DEFINITION
Extraction Blank	A type of negative control to confirm there is no contamination during DNA extractions. Normally an empty filter is extracted and PCR amplified alongside other samples.
No Template Control	A type of negative control during PCR to confirm there is no contamination during the PCR process. Normally nuclease-free water is run in place of DNA on a PCR.

1.4 BACKGROUND

1.4.1 Summary

This protocol describes steps for performing PCR for 18S rRNA V9 marker gene regions using eDNA extracted from Sterivex by USGS. The PCR protocol includes a primary PCR step and a secondary PCR step.

1.4.2 Method description and rationale

This protocol is used for PCR amplifying the 18S rRNA V9 marker gene regions of environmental DNA. It is highly reproducible and can easily be adapted for any number of samples (i.e. a full 96-well plate or a few samples).

1.4.3 Spatial coverage and environment(s) of relevance

This protocol can be used to amplify the 18S rRNA marker gene region of any eDNA sample.

1.5 PERSONNEL REQUIRED

One person with molecular biology experience.

1.5.1 Safety

There are no hazardous chemicals or materials involved in this protocol. Standard lab safety techniques should still be used such as wearing PPE to avoid skin or eye contact.

1.5.2 Training requirements

Basic molecular biology training is sufficient for this protocol including sterile technique, pipetting small volumes and programming/running PCR thermal cyclers.

1.5.3 Time needed to execute the procedure

Protocol takes about 4 hours (240 minutes) including thermal cycler run time.

1.6 EQUIPMENT

For 96-well Plate:

PRODUCT				
NAME AND				
DESCRIPTION	MODEL	MANUFACTURER	QUANTITY	REMARK
Durable equipment				
100-1000 ul Pipette	Eppendorf Research Plus Adjustable-Volume Pipette	Eppendorf	1	Can be substituted with any accurate pipette

PRODUCT NAME AND DESCRIPTIONMODEL		MANUFACTURER	QUANTITY	REMARK
10-100 ul Pipette	Eppendorf Research Plus Adjustable- Volume Pipette	Eppendorf	1	Can be substituted with any accurate pipette
0.1-2.5 ul Pipette	Eppendorf Research Plus Adjustable- Volume Pipette	Eppendorf	1	Can be substituted with any accurate pipette
10-100 ul 8-Channel Pipette	Eppendorf Research Plus 8 Channel Pipette	Eppendorf	1	Can be substituted with any accurate pipette
0.5-10 uL 8-Channel Pipette	Eppendorf Research Plus 8 Channel Pipette	Eppendorf	1	Can be substituted with any accurate pipette
Thermal cycler	Mastercycler Nexus Thermal Cycler	Eppendorf	1	Can be substituted with generic
Consumable equipment				
Gloves	Nitrile Gloves, Exam Grade, Powder-free	ULINE	1	(box) Can be substituted with generic
Kim Wipes	KimWipe Delicate Task Wipers	KimTech	1	(box) Can be substituted with generic
96-well PCR Plate		Armadillo PCR Plate, 96-well, clear, clear wells	ThermoFisher 3	

PRODUCT NAME AND DESCRIPTIONMODEL		MANUFACTURER	QUANTITY	REMARK
PCR Plate Seal	AlumaSeal II Sealing Foils for PCR and Cold Storage	VWR	2	Can be substituted with generic, can use tightly-fitted strip caps in place of seal (box) Can be substituted with generic (boxes) Can be substituted with generic (boxes) Can be substituted with generic (mL)
1000µL Filter Tips	OT-2 Filter Tips, 1000µL	Opentrons	1	
200µL Filter Tips	OT-2 Filter Tips, 200µL	Opentrons	2	
10 ul Filter tips	TipOne Pipette Tips, 10 uL	TipOne	2	
Kapa HiFi Hotstart ReadyMix	Kapa HiFi Hotstart ReadyMix	Roche	0.6	(ul (1uM)) Primer must be diluted from 100uM stocks to 1uM
Molecular water	Invitrogen RT-PCR Grade Water	Fisher Scientific	1 vial	
Forward Primer - 1391F	18S 1391F Fluidigm Primer	IDT	105	
Reverse Primer - EukBR	18S EukBR Fluidigm Primer	IDT	105	
2X Kapa HiFi Hotstart ReadyMix	2X Kapa HiFi Hotstart ReadyMix	Roche	2.4	(mL)

DESCRIPTION	PRODUCT NAME AND MODEL	MANUFACTURER	QUANTITY	REMARK
Index Primer Mix	Illumina Compatible Index Mix (not specified)	missing	0.48	(mL)
Chemicals				
RNase AWAY	RNase AWAY Surface Decontaminant	ThermoFisher Scientific	1	(bottle) Used to sterilize lab surfaces and equipment (wash bottle)
EtOH	Ethanol	Generic Brand	1	Must be molecular grade ethanol (mL)
DI water	Deionized water	Generic	900	(mL)

- Description: E.g., “filter”.
- Product Name and Model: Provide the official name of the product.
- Manufacturer: Provide the name of the manufacturer of the product.
- Quantity: Provide quantities necessary for one application of the standard operating procedure (e.g., number of filters).
- Remark: For example, some of the consumable may need to be sterilized, some commercial solution may need to be diluted or shielded from light during the operating procedure.

1.7 STANDARD OPERATING PROCEDURE

1.7.1 Protocol

1.7.1.1 Preparation

1. Dilute primers from 100 uM to 1 uM.
2. Set up PCR under hood by wiping off all surfaces, pipettes, and racks with RNase AWAY and UV sterilizing for 5-10 mins.
3. Map out order of samples on 96-well PCR plate. Make sure to leave a space for a no template control (NTC) and positive control.

1.7.1.2 PCR

1. Add the following volumes to each well of a 96-well plate:

- 6.25 uL Kapa HiFi Hotstart ReadyMix
- 2.5 ul Fwd primer (1 M) - GTACACACCGCCCGTC
- 2.5 ul Rev primer (1 M) - TGATCCTTCTGCAGGTTCACCTAC

PCR Primer Name	Direction	Sequence (5' -> 3')
1391F	forward	GTACACACCGCCCGTC
EukBr	reverse	TGATCCTTCTGCAGGTTCACCTAC

2. Add 2.0 ul of sample DNA (or molecular water for NTC) to respective wells for a total reaction volume of 13.25 ul per well. Pipette up and down to fully distribute DNA into master mix.
3. Seal plate with PCR plate seal or strip caps.
4. Load plate onto thermal cycler and select program to run the following steps:

PCR step	Temperature	Duration	Repetition
Initial Denaturation	95°C	180s	1x
Denaturation	95°C	30s	30x
Annealing	63°C	30s	30x
Extension	72°C	30s	30x
Final Extension	72°C	5min	1x
Hold	12°C	∞	

1.7.1.3 PCR 2

1. Add the following volumes to each well of a 96-well plate:
 - 25 uL 2X Kapa HiFi Hotstart ReadyMix
 - 5uL Index Primer Mix
2. Add 20 uL of amplified DNA sample to respective wells for a total reaction volume of 50 uL per well. Pipette up and down to fully distribute DNA into master mix.
3. Seal plate with PCR plate seal or strip caps.
4. Load plate onto thermal cycler and select program to run the following steps:

PCR step	Temperature	Duration	Repetition
Initial Denaturation	95°C	180s	1x
Denaturation	95°C	30s	8x
Annealing	55°C	30s	8x
Extension	72°C	30s	8x
Final Extension	72°C	5min	1x
Hold	12°C	∞	

1.7.2 Basic troubleshooting guide

Low Volume Post-PCR

- If using strip-caps, ensure they are tightly fitting on wells. Any gap in the lid will allow for some volume to evaporate during the PCR process on the thermal cycler. If using PCR plate seals, spin down the plate after taking it off the thermal cycler to ensure all condensation is drawn back into the well.

Contamination

- If there are contamination bands appearing on the gel, run another PCR ensuring full sterilization of work spaces and equipment under the hood and use new vials of Kapa HiFi Hotstart ReadyMix and molecular water. If diluted primers are contaminated, use freshly-made aliquot of primers.

1.8 REFERENCES

Not applicable.

1.9 APPENDIX A: DATASHEETS

Not applicable.